

OFFIONG BASSEY EDET

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RESEARCH INTERESTS

Natural Language Processing, Speech Recognition, Speech Synthesis, Machine Translation, Low-Resource Languages, African Language Technology, Multilingual Models, Conversational AI, Language Model Efficiency, and Responsible AI.

EDUCATION

University of Cross River State | 2019 – 2024

B.Sc. Computer Science

PUBLICATIONS & RESEARCH

- Edet, O., et al. (2026). "Towards Digital Preservation of Efik: TTS for Low-Resource African Language" Submitted to **Interspeech 2026**. [Preprint available on request]
- Edet, O., et al. (2026). "Adapting Foundational ASR Models to Efik: An Empirical Study of an Extremely Low-Resource Tonal Language." Accepted to **Speakable** at **LREC 2026**.
- Edet, O., et al. (2026). "Developing an English-Efik Corpus and Machine Translation System for Digitization Inclusion." Accepted to **AfricaNLP** at **EACL 2026**.

EXPERIENCE

Plotweaver | Machine Learning Engineer – NLP | Jan 2026 – Present

- Led the development and deployment of multilingual Text-to-Speech systems for Yoruba, Swahili, Hausa, and other African languages.
- Fine-tuned and evaluated modern TTS architectures including **MMS-TTS, Orpheus-TTS, F5-TTS, and Qwen3-TTS**.
- Built and deployed a **real-time multilingual speech pipeline (ASR → MT → TTS)** for live football commentary and event broadcasting.
- Reduced end-to-end latency from **24 seconds to 3 seconds (87% improvement)** through Faster-Whisper, CTranslate2, and inference optimization.
- Published multilingual speech datasets to **Hugging Face** for reproducible research and internal reuse.
- Deployed scalable inference services and APIs using **AWS SageMaker**.

RapidMind AI (Remote) | *AI Engineer* | Jun 2025 – Oct 2025

- Developed conversational AI systems and Rasa NLU chatbots for fintech customer support automation.

Klas (Remote, Toronto) | *Full Stack Engineer* | May 2024 – Present

- Led development of an **AI Note Taker** for live classes and online sessions.
- Built real-time transcription pipelines for lectures and meetings using speech recognition systems.
- Integrated **LLMs / Generative AI** for automatic lecture summarization, key-point extraction, and structured study notes.
- Improved backend performance and scalability by optimizing server architecture and backend services using Nodejs and Typescript, leading to reduced latency and better traffic management.

- Leveraged AWS Services, including, EC2 Instances, Lambda functions, Dynamo DB, and S3, to optimize image processing, file uploads, and other cloud-based functionalities, contributing to a more efficient and scalable infrastructure.

Sycamore NG / Backend Developer Intern | June 2023 – October 2023

Lagos, GRA, Hybrid

- Built and tested RESTful APIs using Node.js.
- Improved database performance through schema optimization and query tuning.
- Conducted technical product research and reporting.

PROJECTS

Efik Machine Translation System

- Built a high-quality MT system for Efik, a low-resource African language.
- Curated **13,865 parallel English–Efik sentences** with linguists and native speakers.
- Fine-tuned state-of-the-art machine translation models (**mT5-base, M2M-100 and NLLB**) using PyTorch and Hugging Face.

Efik ASR & Text-to-Speech System

- Developed a natural-sounding ASR and TTS system for Efik using Whisper, XLS-R, VITS and diffusion-based models..
- Processed and aligned ~10 hours of speech data..
- Evaluated intelligibility and naturalness using objective and human metrics.

Life Jacket Detection System | 2025

- Developed a real-time object detection system using a fine-tuned YOLOv5 model and OpenCV to enhance maritime safety.
- **Tools:** Python, YOLO, OpenCV, TensorFlow

TECHNICAL SKILLS

Programming Languages: Python, JavaScript, TypeScript, SQL

Machine Learning: PyTorch, TensorFlow, HuggingFace Transformers, Scikit-learn.

Speech AI: Whisper, Faster-Whisper, MMS-TTS, VITS, Orpheus-TTS, ASR, TTS, Qwen3.

NLP / GenAI: NLLB, M2M-100, mT5, Prompt Engineering, RAG, Summarization, Machine Translation.

Cloud & DevOps: AWS SageMaker, EC2, Lambda, S3, Docker, Git, GitHub Actions.

Backend: FastAPI, Node.js, Express.js, REST APIs.

Frontend: React.js, Next.js.

CERTIFICATIONS

- **Deep Learning Specialization**, DeepLearning.AI | 2025
- **Machine Learning**, IBM | 2023
- **Introduction to Software Engineering**, IBM | 2023
- **Python Programming** | University of Michigan – Coursera | 2023